

Milad Heydari

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EDUCATION

Georgia Institute of Technology

BS in Computer Science - AI, Math concentrations

3.86 GPA

MS in Machine Learning

4.0 GPA

Courses: Deep Learning, Deep Reinforcement Learning, Vision-Language Models, Natural Language, OS
ML Systems, HPC & GPU Programming, Computer Vision, Robotics, Statistical ML, Machine Learning

EXPERIENCE

Databricks

Feb 2025 -

Software Engineer — Photon ([paper link](#))

San Francisco, CA

- Improved performance by orders of magnitude on multiple operators. Writing SIMD/CUDA kernels.
- Working primarily with researchers/PhDs on novel architectures. Reading/implementing recent papers

Databricks

May - Aug 2024

Software Engineering Intern — Photon (C++ version of Spark) [mentor](#)

San Francisco, CA

- Proposed novel vectorized broadcast join algorithm that was more efficient than SOTA for sparse batches, setting a benchmarking record (85%+ speedup) & saving many millions in compute to date. O(n) speedup.
- Independently found above optimization opportunity while reading code/paper. Outside of assigned scope, but proposed new project / redesign leveraging memory coalescing and zero-copying to accelerate data loading.

Microsoft

May - Aug 2023

Software Engineering (Applied Science / ML) Intern — Azure Hyperscale Networking [mentor](#)

Redmond, WA

- Led project to model distributed system as real-time updating knowledge graph with devices (nodes), links (edges), and anomaly events (temporal nodes): enabling modeling spatial relationships between anomalies
- Built API that queried network/graph to detect & cluster anomalies. Fed data to LLM using RAG to diagnose issues e.g: using packet loss between datacenters to locate bad cables. 22% drop in avg time to diagnose issues

Cockroach Labs

Jan - May 2023

Software Engineering Intern — Cloud Infrastructure & Distributed Systems

New York, NY

- Shipped service to detect underused Kubernetes clusters and mitigate resource waste by enforcing a resource virtualized 'sleep' state and autoscaling compute based on demand. Azure/AWS/GC. Saving \$70k+ annually
- Enabled multi-cloud data querying/hosting via upgrading (Raft) consensus algorithm for distributed KV layer

Microsoft

May - Aug 2022

Software Engineering Intern — Azure Networking, Applied Science Group under [Jitu Padhye](#)

Redmond, WA

- Built backtesting framework for ML models that allowed A/B and unit testing different models / hyperparams
- Designed architecture/model for time series anomaly detection on streamed data. Performs real time signal processing on a distributed system's network data and triggering mitigation. Prevented 3 incidents in 1st week

Microsoft

May - Aug 2021

Data & Applied Science Intern — Machine Learning Product Integration

Cambridge, MA

- Used auto-encoders to improve model efficiency and preserve privacy of hospital data during federated learning
- Worked with client, Novant Health, to develop custom NLP/ML models and data pipelines to process language data (encoder model) and monitor metrics (RNNs) in diagnoses recommendation/search tool for doctors

Forterra

May - Aug 2020

Robotics & Deep Learning Research Intern — Computer Vision on Autonomous Vehicles

Washington D.C

- Designed novel method of using GANs/programmatic synthetic data generation in training of CV model
- During AV fleet movement, projected position of other vehicles into camera view of each vehicle - using positions to test/calibrate/train predicted positions from onboard vision model and assist with navigation/SLAM

PROJECTS

Distributed Training: locking-free distributed SGD. Scaling Hogwild without atomicity guarantees

Inference Scheduler: vLLM fork with virtual token counter scheduler, batching {prefills, decodes} without stalling

LLM reasoning finetuning: improving Quiet-sTaR & sTaR chain of thought reasoning using DPO training

3D VLM editor: adding/removing objects to Gaussian splats using language. Joint CLIP/grouping training

RL autograd framework: backprop with support for randomized variables and stochastic/RL problems

Robust RLHF in LLM Training: achieving stable reward despite variance in human feedback

SKILLS

Languages: Python, C/C++, Golang, C#, Java, Scala, JavaScript, SQL

Tools: PyTorch, CUDA, Kubernetes, MPI, OpenMP, gRPC, Spark, OpenCV, NumPy, Pandas, HuggingFace, Wandb, LangChain, Cursor, PEFT, Triton, Jax, ROS, Daft